

XXIV IAP Colloquium

Is the IGM temperature- density relation inverted?

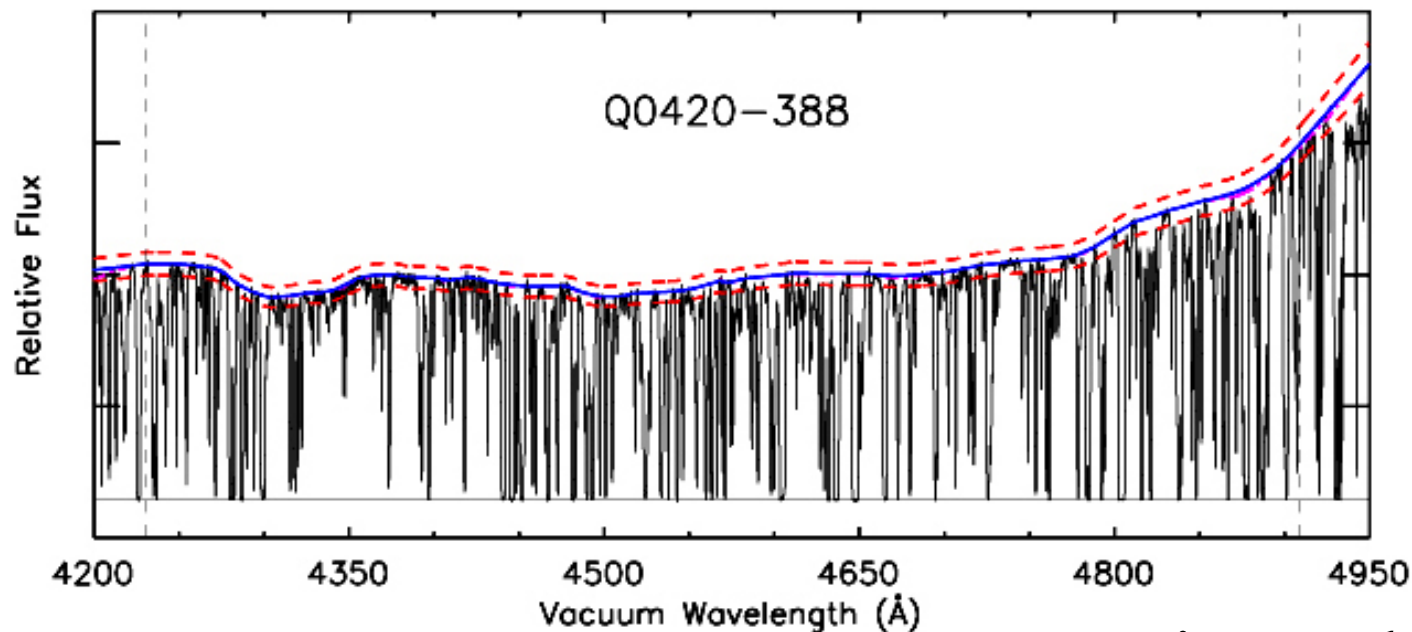
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The observational data



Kim *et al.* (2007)

Set of 18 high resolution VLT/UVES QSO spectra at $1.7 < z < 3.2$ selected from the LUQAS sample (part of ESO LP, **Bergeron *et al.* 2004**). Total path length $\Delta z = 7.83$.

- Detailed analysis and removal of identifiable metal lines.
- Characterise impact of continuum placement and noise properties.
- Result: an improved measurement of the Ly α forest flux distribution.

The Ly α forest and the IGM temperature

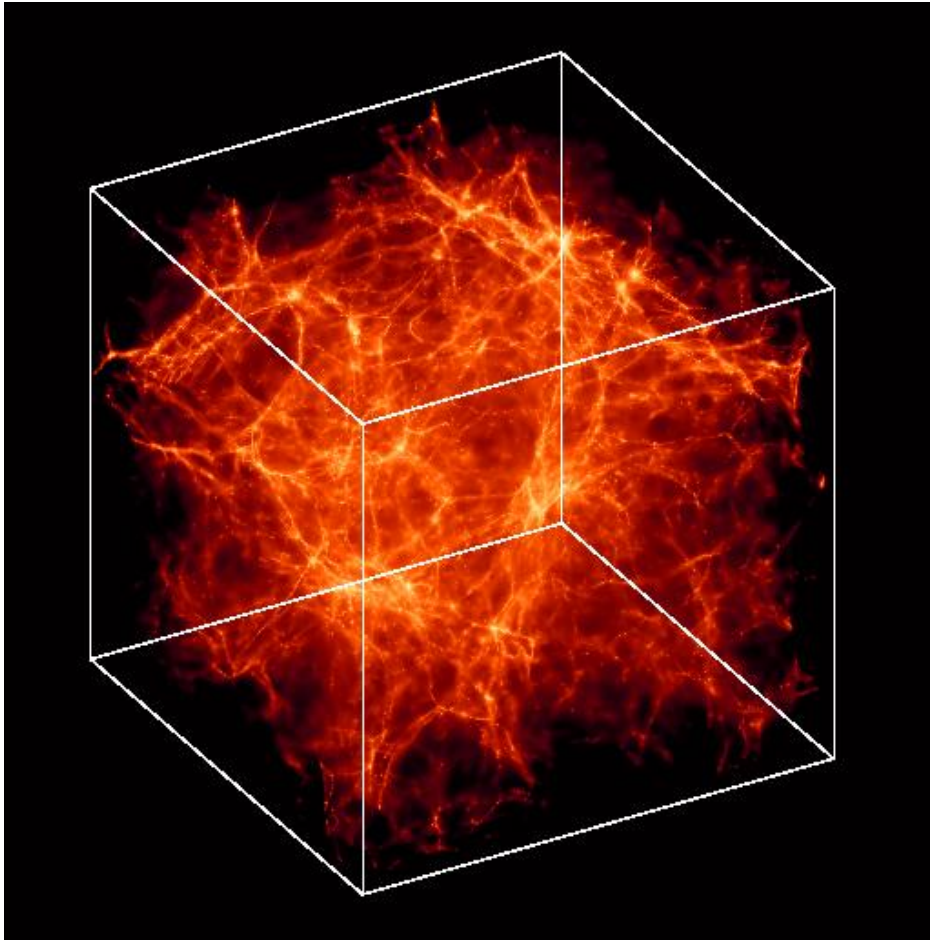
Assuming HI photo-ionisation equilibrium and a temperature-density relation for low density gas, $T=T_0(1+\delta)^{\gamma-1}$ (**Hui & Gnedin 1997**)

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$$

e.g. Rauch et al. (1997)

The Ly α forest opacity is closely linked to the IGM temperature (as well as the UV background, baryon fraction, cosmology...)

Hydrodynamical simulations of the Ly α forest with GADGET-2



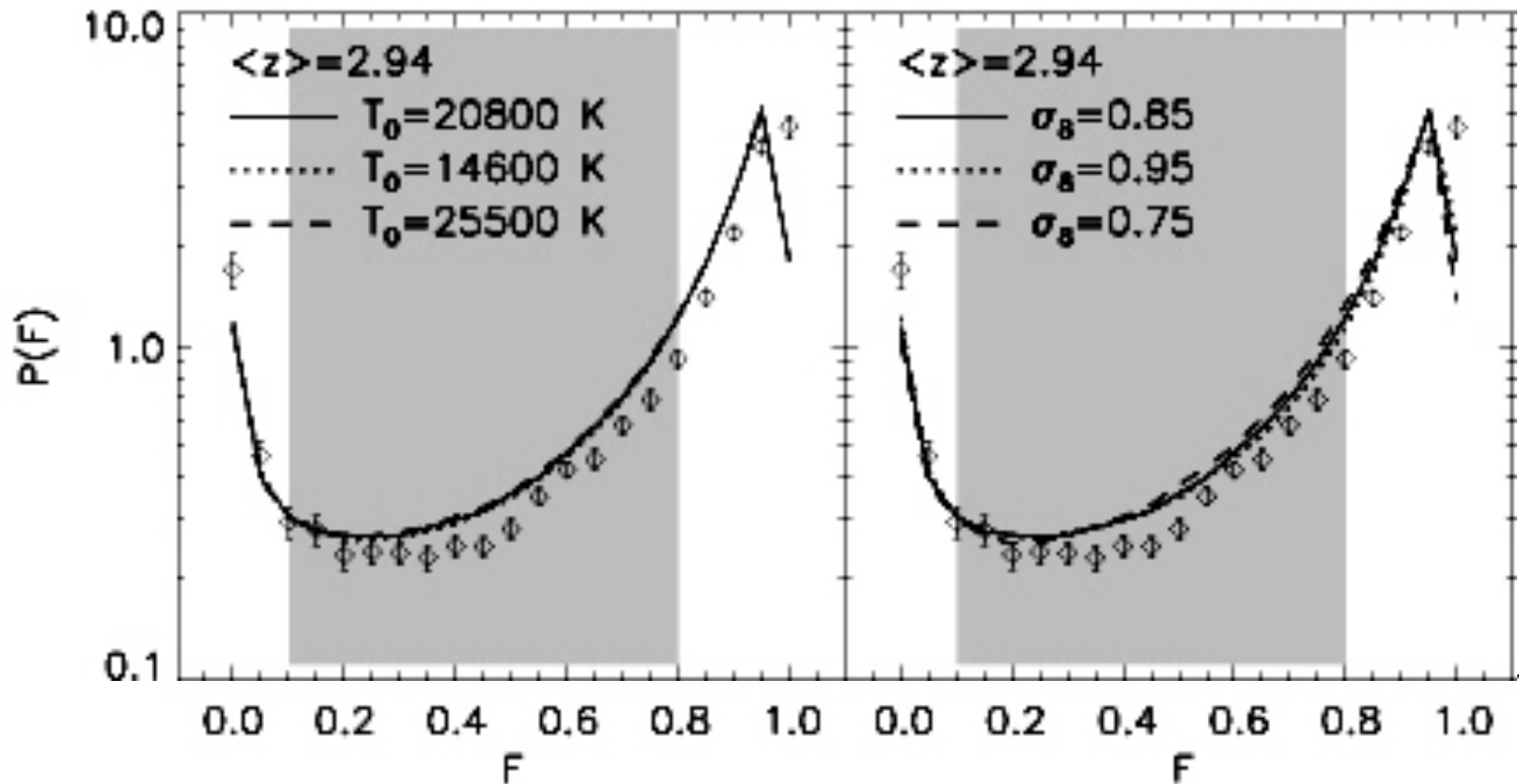
Large set of numerically converged hydrodynamical simulations run with GADGET2 (**Springel 2005**)

WMAP consistent fiducial model

$\Omega_m=0.26$, $\Omega_\Lambda=0.74$,
 $\Omega_b h^2=0.024$, $h=0.72$, $n=0.95$

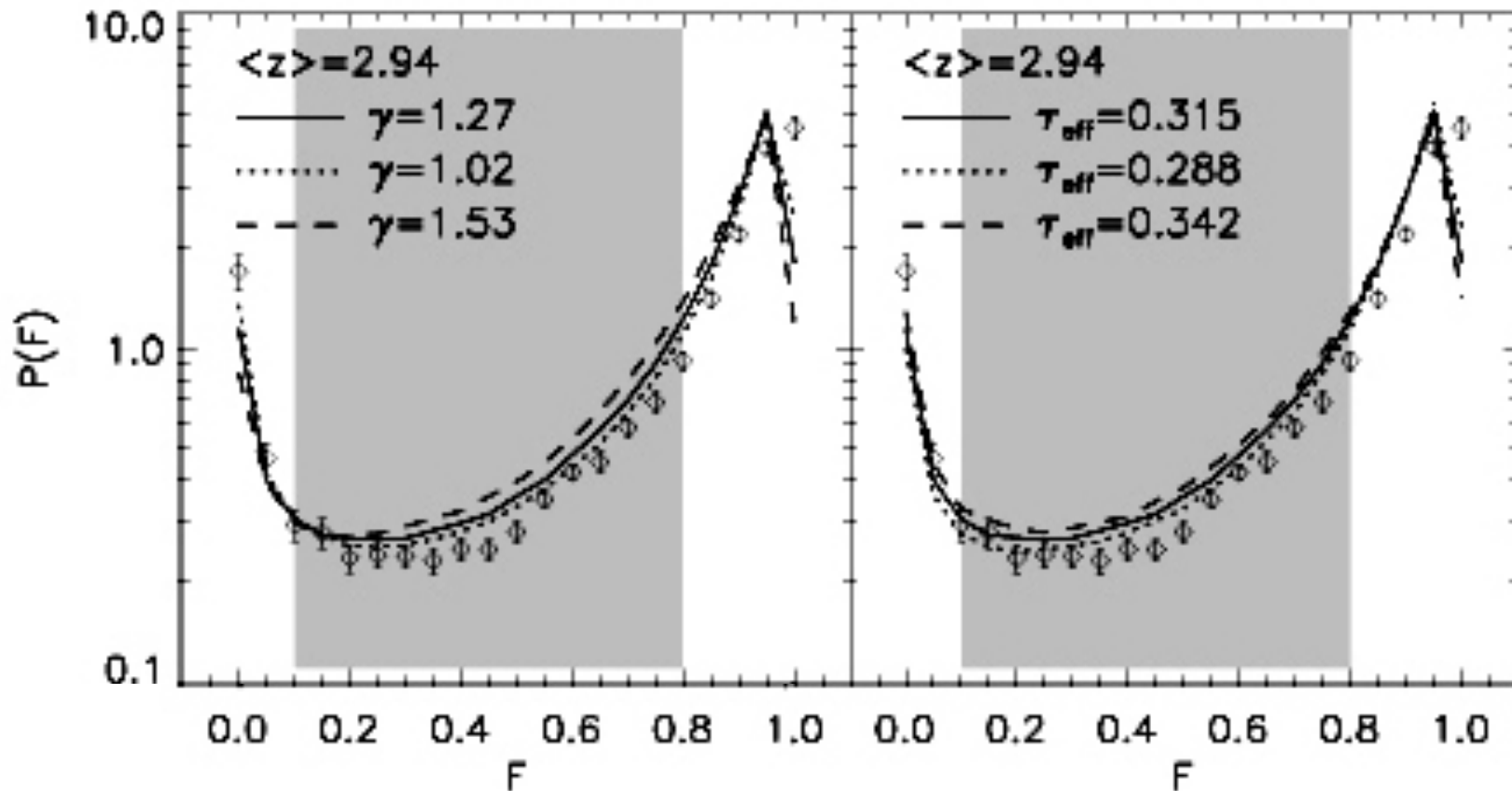
Vary remaining parameters over their observationally established/theoretically expected values and ranges.

Can the standard simulations match the data at $z \sim 3$?



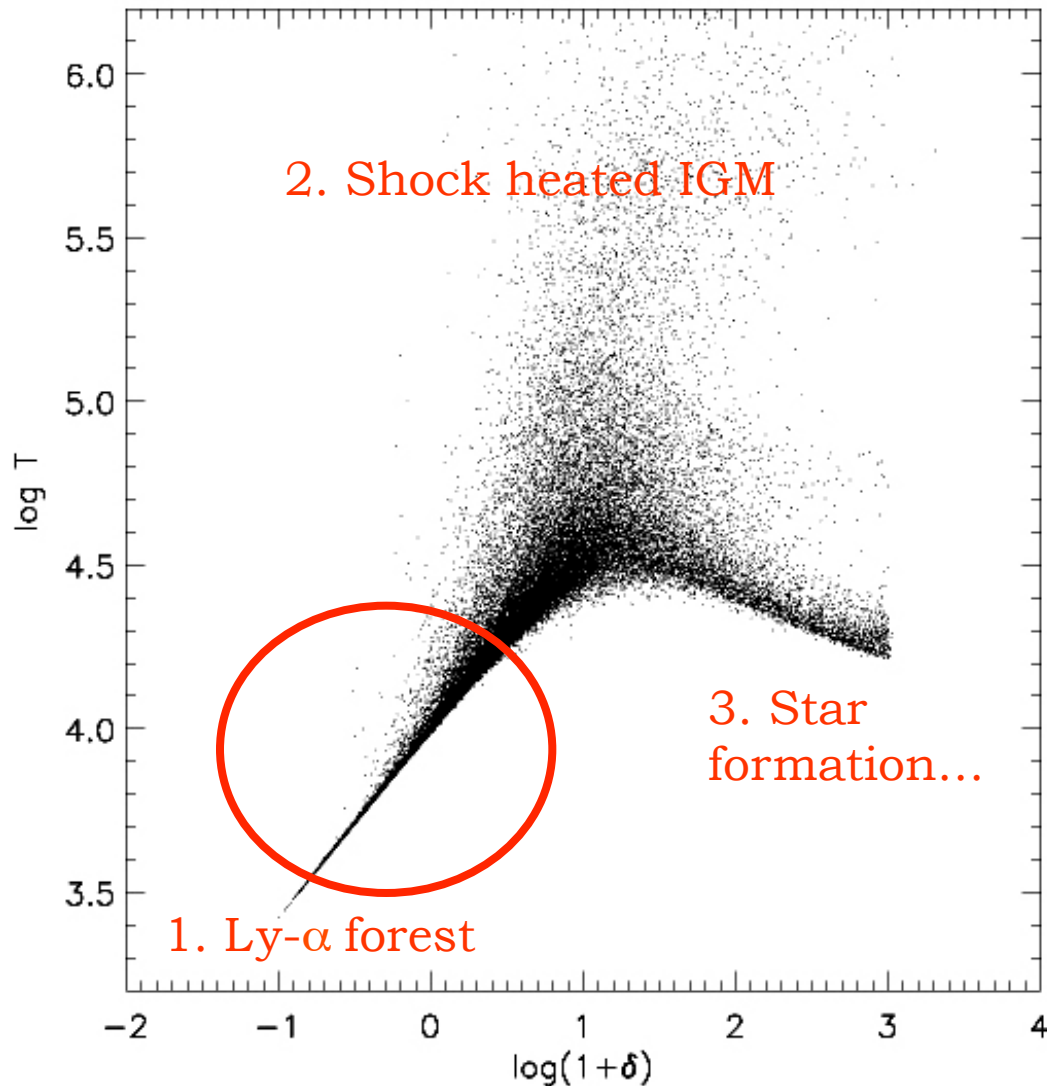
Bolton *et al.* (2008)

Can the standard simulations match the data at $z \sim 3$?



Bolton *et al.* (2008)

The thermal state of the optically thin IGM at $z \sim 3$

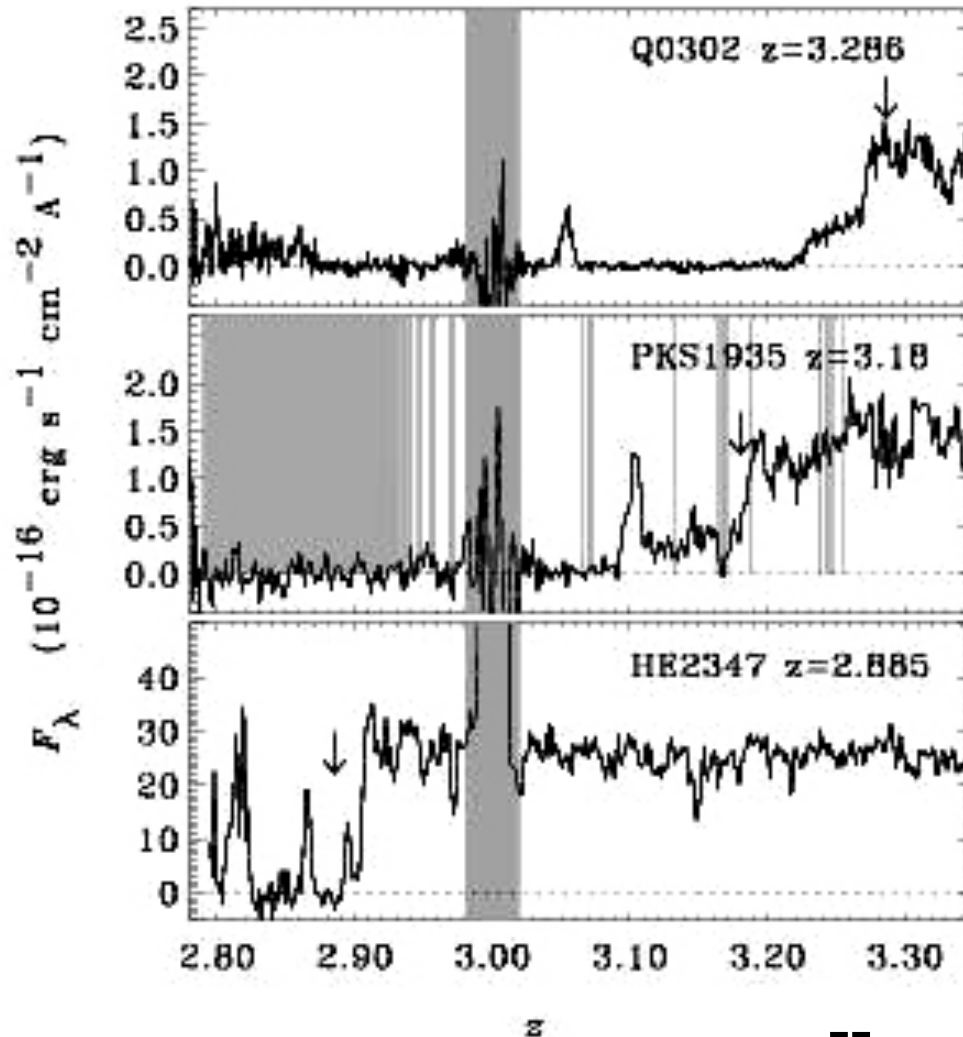


See *e.g.*

Hui & Gnedin (1997)

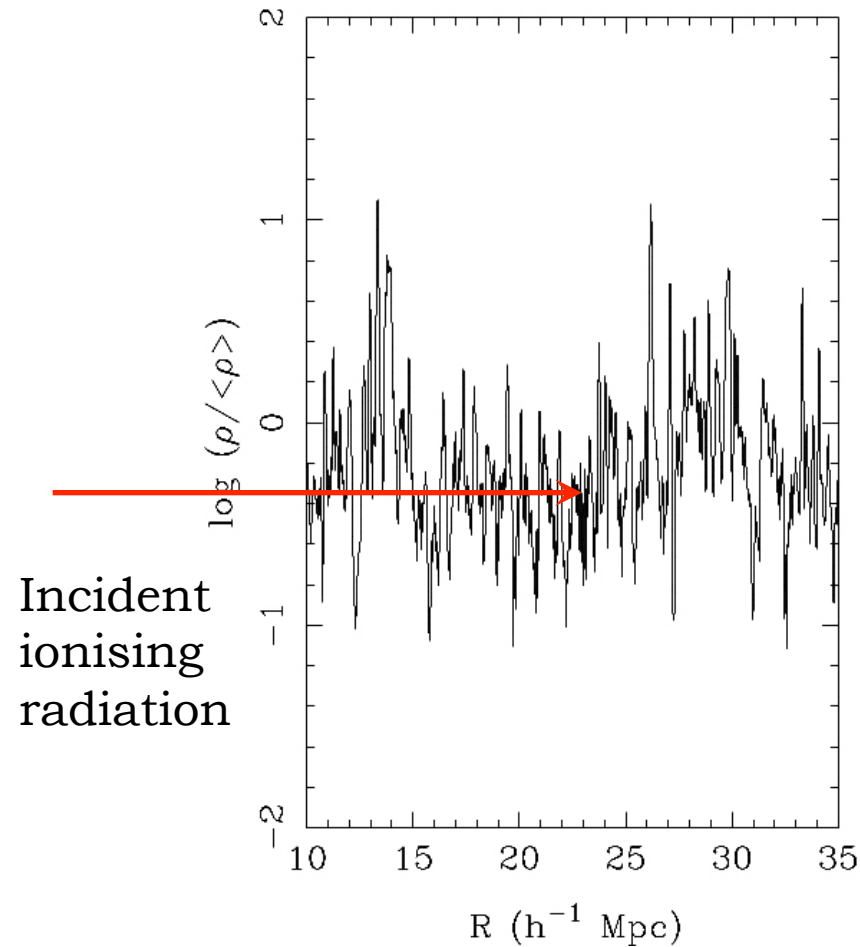
Valageas *et al.* (2002)

The He II Gunn-Peterson trough



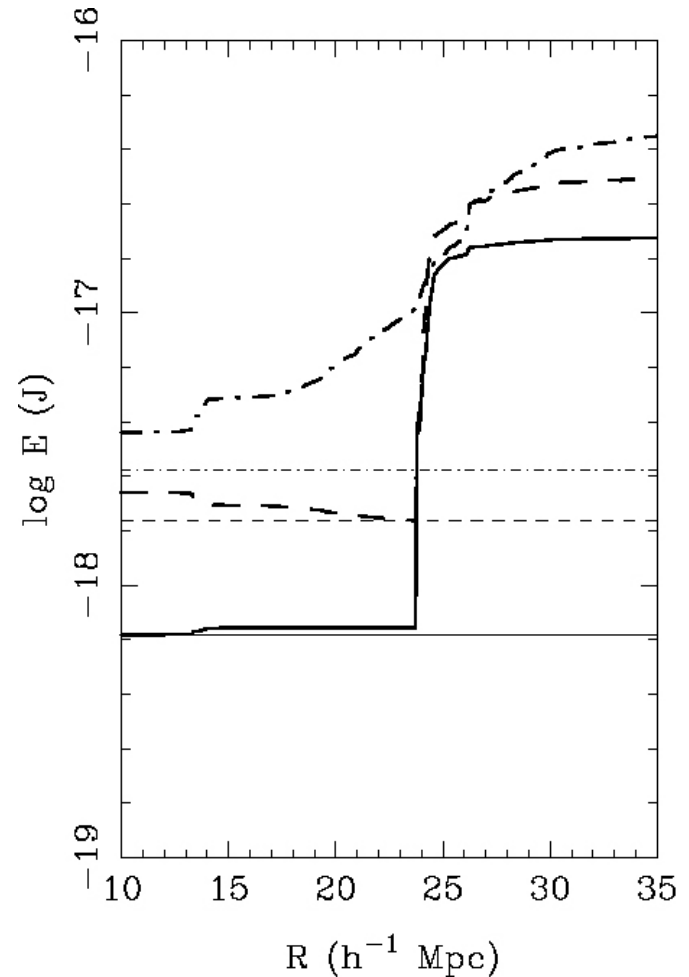
Heap *et al.* (2000)

Radiative transfer effects...



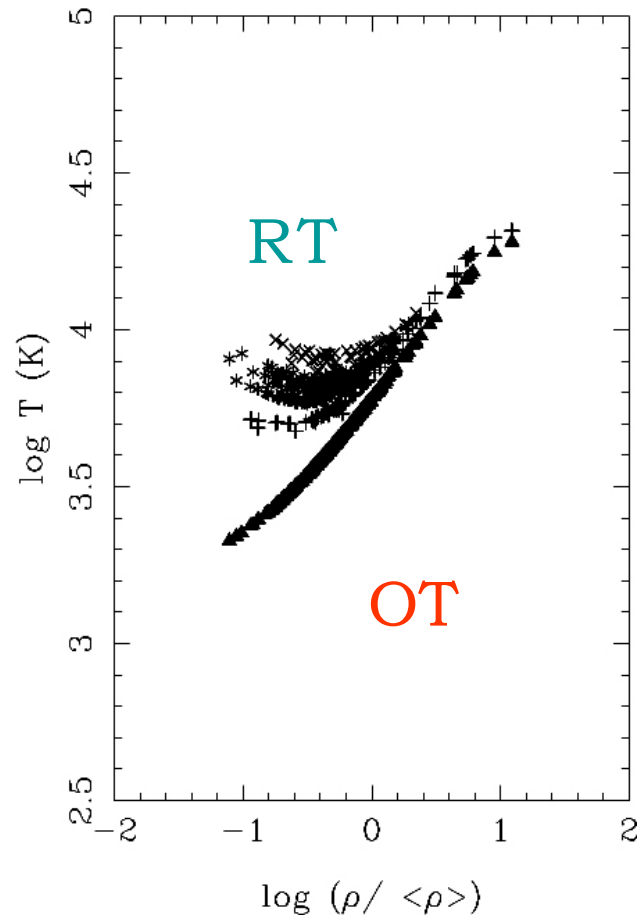
Optically
thick
IGM

$$\sigma_{\nu} \sim \nu^{-3}$$



Bolton, Meiksin & White (2004), see also Abel & Haehnelt (1999)

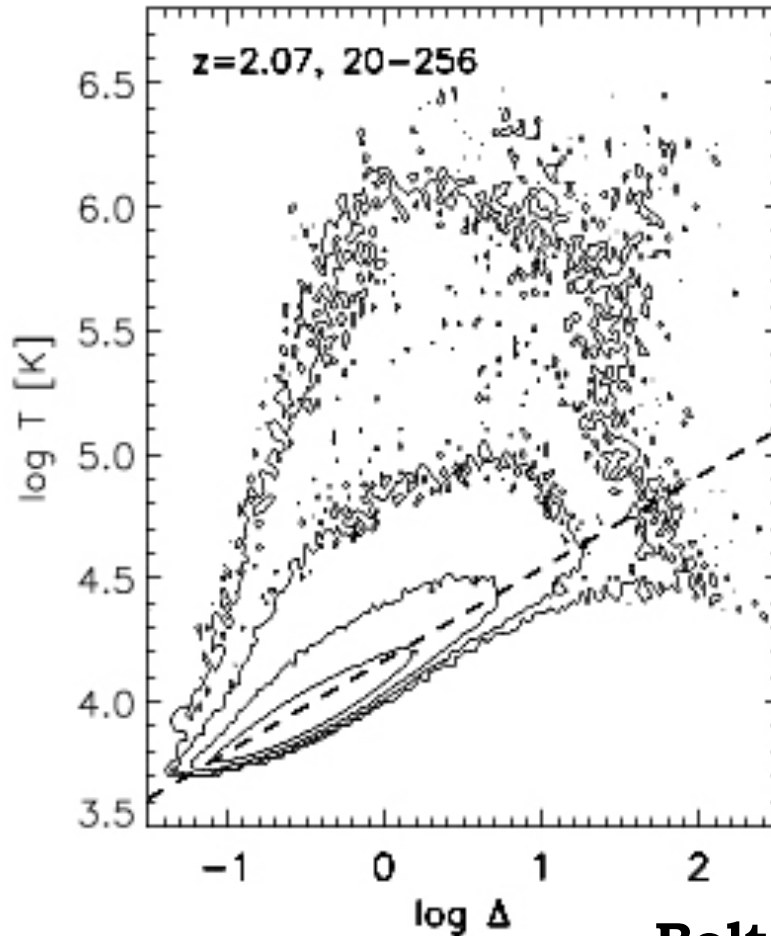
can blur/invert the temperature-density relation.



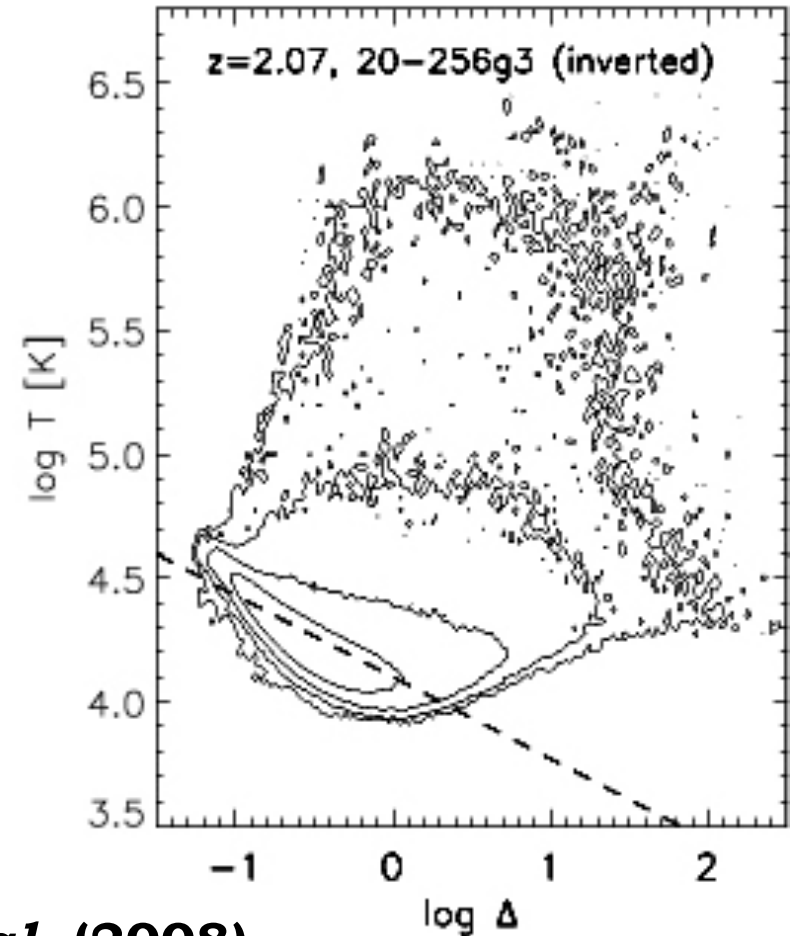
Bolton, Meiksin & White (2004)

Mimicking RT effects on the IGM thermal state

Standard T- ρ relation
(Optically thin, spatially uniform UVB)

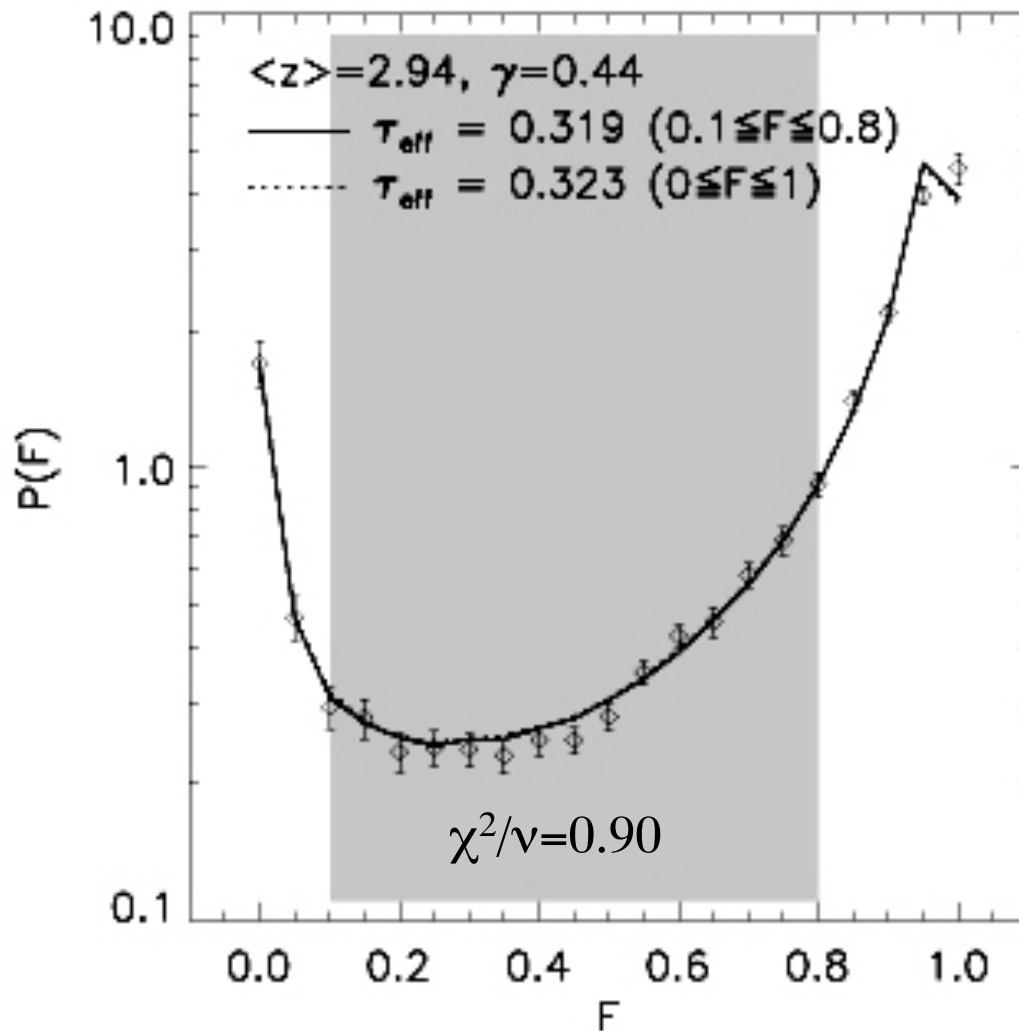


Inverted T- ρ relation
(May mimic He II reionisation effects on the IGM)



Bolton *et al.* (2008)

Evidence for an inverted T- ρ relation?



$$n_{\text{HI}} \Gamma_{\text{HI}} \propto T_0^{-0.7} \Delta^{2-0.7(\gamma-1)}$$

Underdense regions,
associated with flux near
the continuum level
become hotter and more
highly ionised.

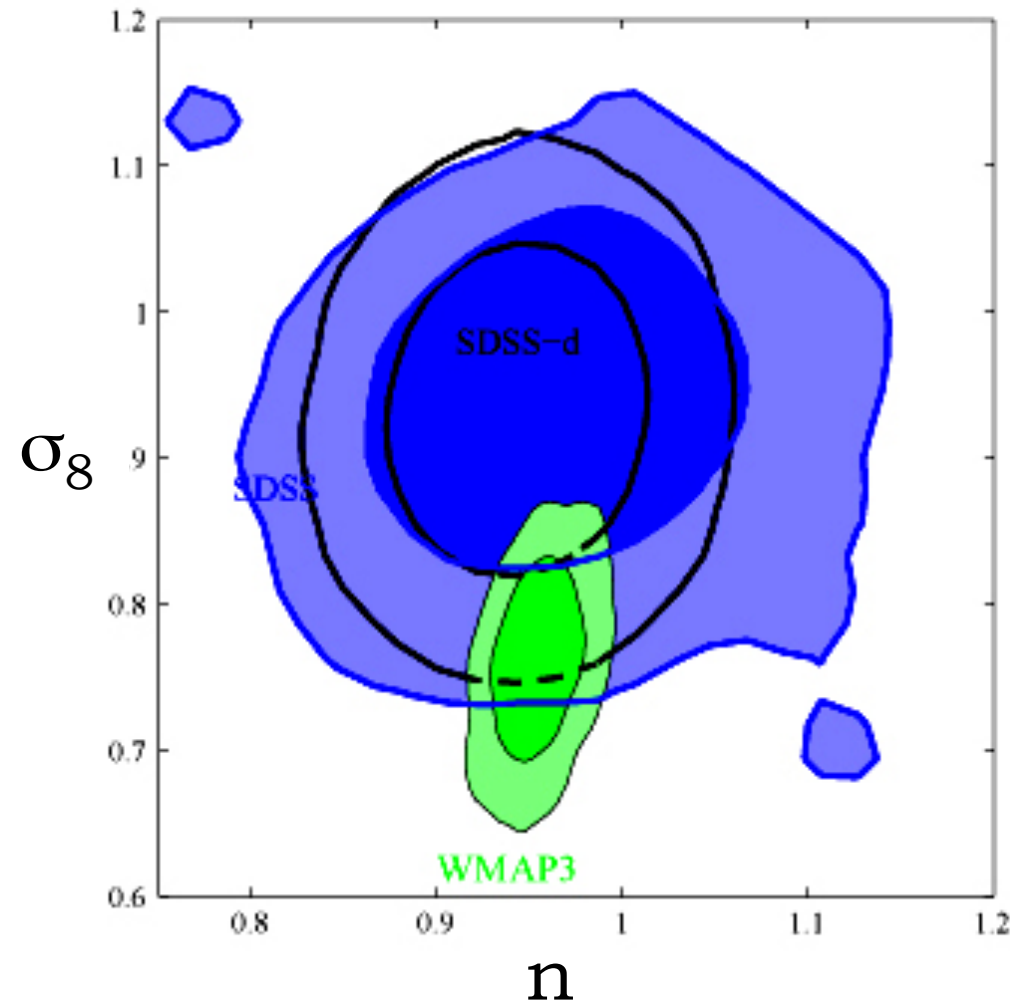
Overdense regions become
cooler and more neutral.

Bolton *et al.* (2008)

and see also

Becker *et al.* (2007)

Systematic uncertainties in modelling the flux power spectrum?



WMAP 3

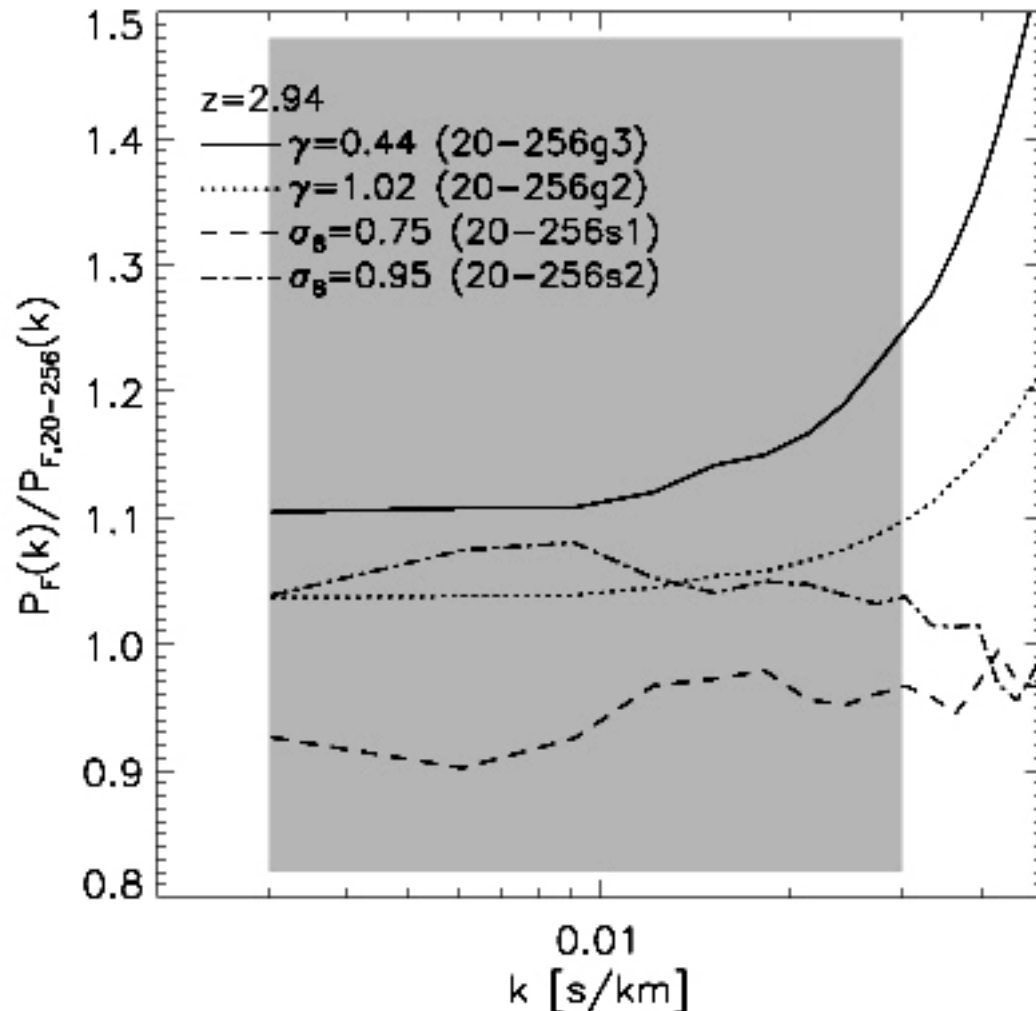
$$\sigma_8 = 0.76 \pm 0.05, n = 0.96 \pm 0.02$$

WMAP 3 + SDSS Ly- α

$$\sigma_8 = 0.86 \pm 0.03, n = 0.96 \pm 0.02$$

Viel, Haehnelt & Lewis (2006)

Implications for cosmology with the Ly α forest



Bolton *et al.* (2008)

Inverted γ increases power at the scales relevant for measuring the matter power spectrum

Colder, more neutral material in the overdense regions associated with the most prominent lines increases the fluctuations in the transmission.

Acts in the same way as a larger σ_8 !

Conclusions

- Is the IGM temperature-density relation inverted? Perhaps, but the real situation may be somewhat more complex.
- The data are indeed consistent with voids in the IGM which are significantly hotter than usually assumed (**Bolton *et al.* 2008**).
- Radiative transfer effects during He II reionisation by quasars *may* provide the missing physics which improves the agreement of simulations with the data.
- This also has important implications for cosmological parameters derived from the Ly α forest. Hotter voids can mimic the effect of larger σ_8
- Further investigation is needed - more detailed simulations of HeII reionisation and a full analysis of other statistics are required...